

Electron configurations

Learning outcomes

By the end of this section you should be able to apply the Aufbau principle, Hund's rule and the Pauli exclusion principle to deduce electron configurations for atoms and ions up to $Z = 36$.

When sending a letter or parcel, it is necessary to write the address to where it is to be delivered. An address usually has the postal code, city name, as well as the street name and house name or number. Although we can't know exactly where electrons are located in an atom, we can write an electron configuration for an atom. This is similar to an address and tells us where the electrons are located – in which main energy level and sublevel – as well as how many electrons there are in that particular region. In this section, you will learn how to write and interpret electron configurations for atoms and ions.

Electron configurations

An atomic orbital is a region of space inside the atom where there is a high probability of finding an electron. In section S1.3.3–4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>) you learned the shapes and s and p atomic orbitals. Recall that there are also d and f orbitals but the shapes of these are not required.

To fill atomic orbitals with electrons, we must follow certain principles, which are summarised here.

- The Aufbau principle states that when adding electrons to an atom, the lower energy orbitals must be filled first.
- The Pauli exclusion principle states that an atomic orbital can only hold two electrons and they must have opposite spins.
- Hund's rule states that when we have degenerate orbitals (orbitals of the same energy) then each orbital is filled with a single electron before being doubly occupied.

For example, a helium atom contains two electrons in the first energy level in an s orbital – this electron configuration would be stated as $1s^2$. The '1' represents the energy level, 's' the shape of the orbital and '2' the number of electrons. You will learn how to state the electron configurations for atoms with more electrons below.

Energy level diagrams

Within a given main energy level, s orbitals are of lower energy than p orbitals which are of lower energy than d orbitals. f orbitals have the highest energy. So the order of energy of the sublevels is $s < p < d < f$. **Figure 1** shows the energy of the 4s, 4p, 4d and 4f orbitals in the $n = 4$ energy level.

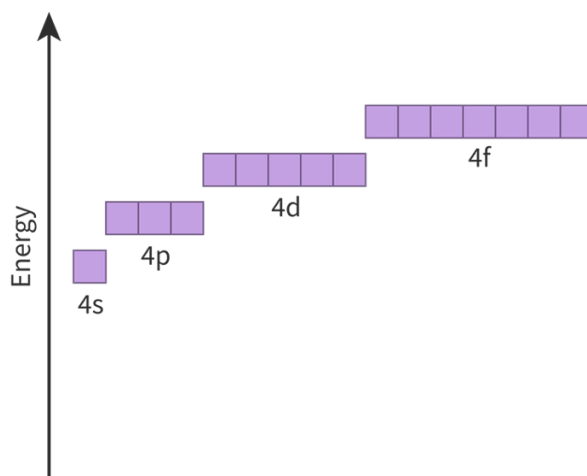


Figure 1. The 4s, 4p, 4d and 4f atomic orbitals in the $n = 4$ energy level.

 More information for figure 1

The image is a diagram illustrating the energy levels of atomic orbitals in the $n=4$ energy level. It shows a stepped arrangement, with each step representing an orbital sublevel: 4s, 4p, 4d, and 4f. The 4s orbital is depicted at the lowest energy level, followed by 4p, 4d, and 4f, which is at the highest level. Each orbital is represented by rectangles aligned on horizontal planes, indicating the relative energy and inclusion of electron occupancy levels.

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The order of the sublevels from 1s to 6s in terms of their energy is shown in **Figure 2**.

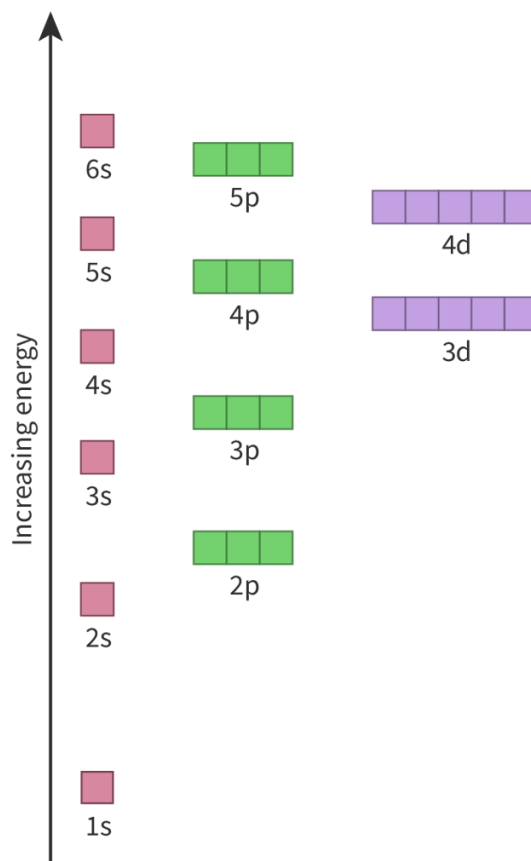


Figure 2. The 1s to 6s atomic orbitals in order of increasing energy.

 More information for figure 2

The image is a diagram depicting the order of atomic orbitals from 1s to 6s in terms of increasing energy. On the left, there is a vertical arrow labeled 'Increasing energy'. Next to the arrow, boxes representing the s, p, and d orbitals are arranged in columns. The 1s orbital is at the bottom, followed by 2s, 2p, 3s, 3p, 3d, 4s, 4p, 4d, 5s, 5p, and finally 6s at the top. Each of these orbitals is illustrated by a series of small boxes, where s orbitals have a single box, p orbitals have three boxes, and d orbitals have five boxes, suggesting the number of electrons each can hold. The diagram visually represents the relative energy levels of these atomic orbitals as they ascend the chart.

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Here are some important points to note about the energy of the sublevels.

- The atomic orbitals within the same sublevel are of equal energy, known as degenerate orbitals. For example, the three p orbitals in the 2p sublevel are of equal energy. This also applies to the three p orbitals in the 3p sublevel and the five d orbitals in the 3d sublevel, etc.

- There is an overlap in energy between the 3d and 4s sublevels. The 4s sublevel is of lower energy and fills before the 3d sublevel*. This has important consequences when writing the electron configuration of the d-block elements.

*The above paragraph states that the 4s sublevel fills before the 3d sublevel, but this is not the case for all atoms. The reasons for this are beyond SL and HL chemistry so will not be covered in this text. Students should be aware that the 4s sublevel fills before the 3d sublevel and electrons are removed from the 4s sublevel first when forming ions. If you are interested in reading more about this phenomenon, you can do so [here](https://ericscerri.blogspot.com/2012/06/trouble-with-using-aufbau-to-find.html) (<https://ericscerri.blogspot.com/2012/06/trouble-with-using-aufbau-to-find.html>).

The order of filling of the sublevels is summarised in **Figure 3**. This can be learned to help you remember the order of filling. Alternatively, another method can be used, known as the block method. This is covered in [subtopic S3.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>).

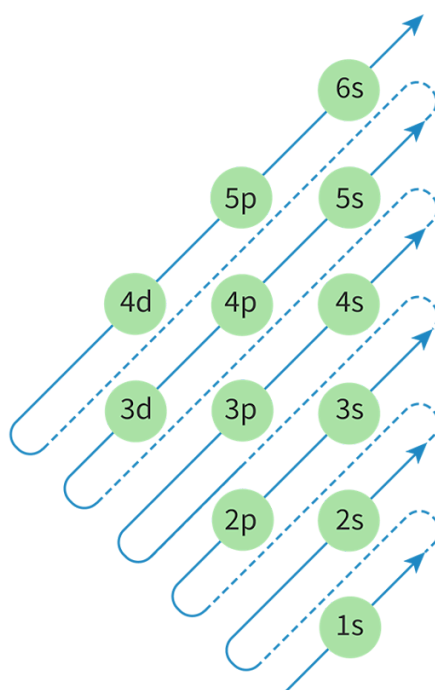


Figure 3. The order of filling of the sublevels 1s through to 6s.

 More information for figure 3

The diagram illustrates the order of filling electron sublevels in an atom. The layout includes sublevels labeled as 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, and 6s. Each sublevel is represented in a separate section with a label, and arrows indicate the sequence in which electrons fill these sublevels, starting from 1s in the bottom right to 6s at the top right.

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Test your understanding by trying **Interactive 1**. Drag and drop the components in the correct order to make the electron configuration of selenium, Se.

2p⁶

4s²

1s²

3s²

2s²

3d¹⁰

3p⁶

4p⁴

|

☐ Check

Interactive 1. Electron Configuration of Selenium, Se.

Making connections

In section S3.1.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrons-and-the-periodic-table-id-44056/>) you will learn an alternative method to determine the electron configuration of an element using the periodic table. You can use whichever method you find easier as both give the correct electron configurations.

Figure 4 shows the notation used when writing electron configurations. The number in front of the letter is the principal quantum number, n , which gives the number of the main energy level. The letter refers to the sublevel (s, p, d, f) and the number in superscript gives the number of electrons in the sublevel.

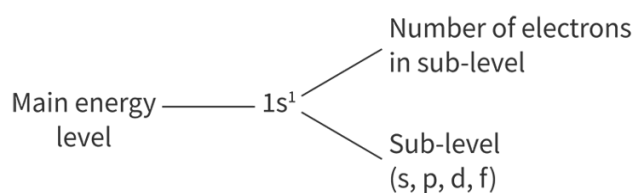


Figure 4. The notation used for electron configurations.

More information for figure 4

This is a diagram illustrating the notation used for writing electron configurations. It outlines the principal quantum numbers represented by numbers, which denote the main energy levels: 1, 2, 3, 4, etc. The diagram shows the sublevels as letters: s, p, d, f,

corresponding to different shapes or types of electron orbitals. Accompanying these letters, there are superscript numbers that denote the number of electrons within each sublevel. The flow of the diagram progresses from the lowest energy level to the highest, demonstrating a systematic buildup of electrons according to quantum mechanics' principles. Key segments of the diagram include the layout of energy levels vertically, the breakdown of each level into sublevels, and the electron count in superscript. This visual representation helps in understanding the electronic configuration of elements.

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Worked example 1

Write the full electron configurations for the first ten elements of the periodic table, from hydrogen to neon.

Starting with hydrogen, H, which is atomic number 1. Atoms have the same number of protons and electrons, so hydrogen has 1 electron.

According to the Aufbau principle we add electrons to the lowest energy orbitals first, so we put one electron in the 1s orbital. The electron configuration of hydrogen is:



Next is helium, He, with the atomic number 2 which means helium has 2 electrons. This gives the electron configuration:



Next is lithium, Li, with atomic number 3 meaning it has 3 protons and 3 electrons. The 1s orbital can only hold two electrons so we need to start filling the 2s sublevel. The electron configuration of lithium is:

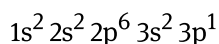


From here on, we will just add one to the atomic number and one to the number of electrons. We will also follow the Aufbau principle of filling lowest energy orbitals first.

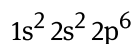
Element	Atomic number	Ele
Beryllium, Be	4	
Boron, B	5	

Element	Atomic number	Ele
Carbon, C	6	
Nitrogen, N	7	
Oxygen, O	8	
Fluorine, F	9	
Neon, Ne	10	

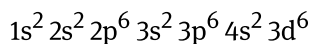
When writing the electron configurations of ions, remember to either add the correct number of electrons (according to the charge) for negative ions or remove the correct number of electrons for positive ions. Note that electrons are always removed from higher energy orbitals first. For example, aluminium, which forms a 3+ ion, Al^{3+} , has the electron configuration:



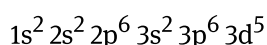
From which orbital would the first electron be removed? The first electron is removed from the 3p orbital because it is the highest energy orbital. The next two electrons are removed from the 3s orbital to give a total of three electrons removed. Therefore, the electron configuration of the Al^{3+} ion is:



For the transition elements, it is important to remember that electrons are removed from the 4s orbital first, and then from the 3d sublevel. For example, iron, Fe has the electron configuration:



The iron atom can lose three electrons to form the Fe^{3+} ion. Because the 4s orbitals have higher energy than the 3d orbitals, the two 4s electrons are removed first. The next electron is lost from the 3d sublevel, so the electron configuration of the Fe^{3+} ion is:



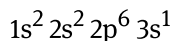
Worked example 2

Write the full electron configurations for the following ions.

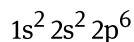
- (a) Na^+
- (b) Mg^{2+}
- (c) F^-
- (d) O^{2-}

- (e) Ni^{2+}

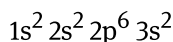
(a) Start by writing the electron configuration for the sodium atom.



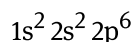
Na^+ is a cation with a 1+ charge meaning that it has one less electron than protons. We remove outer electrons first so the electron configuration of Na^+ is:



(b) Following the same method for Mg^{2+} , we start with the full electron configuration of the magnesium atom:

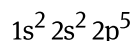


Because it is a cation with a 2+ charge, we then remove the two outer electrons in the 3s sublevel. The electron configuration of Mg^{2+} is:

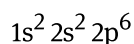


Note that the electron configurations for Na^+ and Mg^{2+} are the same; known as being isoelectronic (have the same electron configuration).

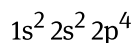
(c) F^- is an anion so we need to add electrons to the electron configuration of the fluorine atom, which is:



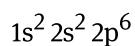
The F^- ion has a 1- charge so we add one electron to the 2p sublevel:



(d) O^{2-} is an anion so we need to add electrons to the electron configuration of the oxygen atom, which is:

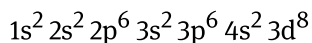


The O^{2-} ion has a 2- charge so we add two electrons to the 2p sublevel:

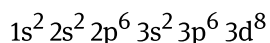


Once again you will notice that all of the ions in the example are also isoelectronic.

(e) The electrons configuration of the nickel atom is:

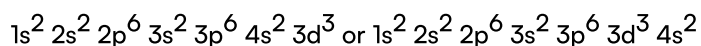


The Ni^{2+} ion has a 2+ charge so two electrons are removed from the 4s sublevel to give the electron configuration:



Study skills

Although the 4s sublevel fills before the 3d sublevel, the electron configuration can be written with the 4s before, or after, the 3d sublevel. For example, the electron configuration of vanadium, V, can be written as:



Both ways of writing are accepted as being correct.

For atoms or ions with higher atomic numbers, we can write the condensed electron configurations. This notation uses the symbol of a noble gas to represent the core electrons. **Table 1** shows the noble gas symbol and the core electrons represented by that symbol.

Table 1. Noble gas symbols and the core electrons represented by the symbols.

Symbol	Core electrons
[He]	$1s^2$
[Ne]	$1s^2 2s^2 2p^6$
[Ar]	$1s^2 2s^2 2p^6 3s^2 3p^6$

Worked example 3

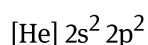
Write the condensed electron configurations for the following atoms.

- (a) Carbon, C
- (b) Phosphorus, P
- (c) Potassium, K
- (d) Cobalt, Co
- (e) Bromine, Br

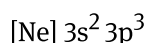
For each example, we will start by writing the full electron configuration and then write the condensed electron configuration using the noble gas symbol to represent the core electrons.

(a) For carbon, C, the full electron configuration is $1s^2 2s^2 2p^2$

The core electrons are the electrons in the 1s sublevel so the noble gas symbol is [He]. The condensed electron configuration is:



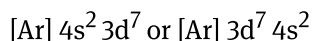
(b) For phosphorus, P, the full electron configuration is $1s^2 2s^2 2p^6 3s^2 3p^3$. The core electrons are the electrons in the 1s, 2s, and 2p sublevels so the noble gas symbol is [Ne]. The condensed electron configuration is:



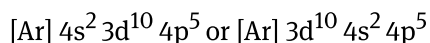
(c) For potassium, K, the full electron configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$. The core electrons are the electrons in the 1s, 2s, 2p, 3s and 3p sublevels so the noble gas symbol is [Ar]. The condensed electron configuration is:



(d) For cobalt, Co, the full electron configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$. The core electrons are the electrons in the 1s, 2s, 2p, 3s and 3p sublevels so the noble gas symbol is [Ar]. The condensed electron configuration is:

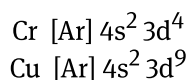


(e) For bromine, Br, the full electron configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$. The core electrons are the electrons in the 1s, 2s, 2p, 3s and 3p sublevels so the noble gas symbol is [Ar]. The condensed electron configuration is:



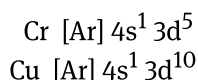
Exceptions to the Aufbau principle

There are two important exceptions to the Aufbau principle that you should be aware of. These are for the elements copper (symbol Cu) and chromium (symbol Cr). Following the order of filling according to the Aufbau principle you would *expect* the electron configurations to be as follows:



However, the observed configurations are different from what is expected.

The actual electron configurations for copper and chromium are:



For chromium, there is one electron in the 4s sublevel and five in the 3d sublevel. And for copper there is one electron in the 4s sublevel and ten electrons in the 3d sublevel. You should take care to remember these exceptions, but no explanation is required.

Theory of Knowledge

Rules and patterns, such as Hund's rule, the Aufbau principle and the Pauli exclusion principle are used to fill atomic orbitals with electrons for elements. However, rules often have exceptions, as you learned above with the examples of chromium and copper. Several other exceptions for electron configuration also occur in elements containing electrons in the d and f sublevels. Do exceptions make a rule less valid?

Orbital diagrams

Electrons in atomic orbitals can be represented by orbital diagrams, also known as arrow in-box diagrams. These diagrams use boxes to represent atomic orbitals and single-headed arrows to represent electrons. According to the Pauli exclusion principle we can only put two arrows in a box, and they must have opposite spins – this is represented by one arrow pointing up and one arrow pointing down. For example, the electron configuration of nitrogen, N, is $1s^2 2s^2 2p^3$. The arrow in box diagram for nitrogen is shown in **Figure 5**. Note that it is important how we place the arrows (electrons) in the boxes (orbitals) as will be discussed next.

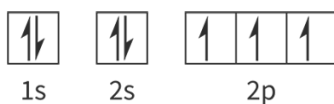


Figure 5. The orbital diagram for nitrogen, N.

 More information for figure 5

The image represents an orbital diagram for nitrogen (N). It consists of a series of boxes and arrows, indicating the arrangement of electrons within nitrogen's atomic orbitals according to the arrow-in-box format. The configuration begins with the 1s orbital, which contains two arrows indicating two electrons with opposite spins, one pointing upwards and one downwards. This is followed by the 2s orbital, also containing two electrons with opposite spins. Following are three 2p orbitals, each with a single arrow pointing upwards, representing nitrogen's three unpaired electrons according to Hund's rule.

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In the 2p sublevel you can see that we put one electron in each 2p atomic orbital according to Hund's rule. This states that degenerate orbitals are singly occupied before being doubly occupied. The next element, oxygen, has the electron configuration $1s^2 2s^2 2p^4$. The orbital diagram of oxygen, O, is shown in **Figure 6**. From this you can see that once each orbital has a single electron in it, we then start doubly occupying each orbital.

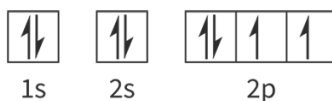


Figure 6. The orbital diagram for oxygen, O.

 More information for figure 6

The image is an orbital diagram for oxygen representing its electron configuration. It shows that the oxygen atom has three main sections for sublevels: 1s, 2s, and 2p. The 1s and 2s sublevels each have a pair of dots indicating that they are fully occupied with two electrons. The 2p sublevel has three parts, each represented by a box. One electron occupies each of the three boxes, following Hund's rule, and then one of these boxes has a pair of electrons, illustrating the electron configuration $1s^2 2s^2 2p^4$. This arrangement shows that the last electron is added to one of the already singly-occupied 2p orbitals, resulting in one full box in the 2p sublevel and two half-full boxes, matching the electron count of an oxygen atom.

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Making connections

What is the relationship between energy sublevels and the block nature of the periodic table? In [subtopic S3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/) you will learn about the relationship between the number of occupied energy levels in an atom and its position in the periodic table.

In this activity you will practise writing electron configurations.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Reflecting at all stages of the assessment and learning cycle
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

In this activity, you will practise writing electron configurations for atoms. Write full and condensed electron configurations for the following atoms.

Atoms	
1. He	11. Ar
2. Li	12. Ca
3. B	13. Ti
4. C	14. Mn
5. O	15. Ni
6. Ne	16. Zn
7. Na	17. Ge
8. Al	18. Se
9. P	19. Br
10. Cl	20. Kr

Draw orbital diagrams for the following atoms.

- (a) Carbon
- (b) Nitrogen
- (c) Oxygen

Electron Configuration	
1. He: $1s^2$	11. Ar: $1s^2 2s^2 2p^6 3s^2$ or [Ne] $3s^2 3p^6$
2. Li: $1s^2 2s^1$ or [He] $2s^1$	12. Ca: $1s^2 2s^2 2p^6 3s^2$ or [Ar] $4s^2$
3. B: $1s^2 2s^2 2p^1$ or [He] $2s^2 2p^1$	13. Ti: $1s^2 2s^2 2p^6 3s^2$ or [Ar] $4s^2 3d^2$
4. C: $1s^2 2s^2 2p^2$ or [He] $2s^2 2p^2$	14. Mn: $1s^2 2s^2 2p^6 3s^2$ or [Ar] $4s^2 3d^5$
5. O: $1s^2 2s^2 2p^4$ or [He] $2s^2 2p^4$	15. Ni: $1s^2 2s^2 2p^6 3s^2$ or [Ar] $4s^2 3d^8$
6. Ne: $1s^2 2s^2 2p^6$ or [He] $2s^2 2p^6$	16. Zn: $1s^2 2s^2 2p^6 3s^2$ or [Ar] $4s^2 3d^{10}$
7. Na: $1s^2 2s^2 2p^6 3s^1$ or [Ne] $3s^1$	17. Ge: $1s^2 2s^2 2p^6 3s^2$ or [Ar] $4s^2 3d^{10} 4p^2$
8. Al: $1s^2 2s^2 2p^6 3s^2 3p^1$ or [Ne] $3s^2 3p^1$	18. Se: $1s^2 2s^2 2p^6 3s^2$ or [Ar] $4s^2 3d^{10} 4p^4$
9. P: $1s^2 2s^2 2p^6 3s^2 3p^3$ or [Ne] $3s^2 3p^3$	19. Br: $1s^2 2s^2 2p^6 3s^2$ or [Ar] $4s^2 3d^{10} 4p^5$
10. Cl: $1s^2 2s^2 2p^6 3s^2 3p^5$ or [Ne] $3s^2 3p^5$	20. Kr: $1s^2 2s^2 2p^6 3s^2$ or [Ar] $4s^2 3d^{10} 4p^6$

